

EDS 212: Day 3, Lecture 2

Vectors and matrices

August 5th, 2025

Summation Notation

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Summation notation

Summation notation is a compact way to write the sum of a sequence of values.

The idea is that, instead of writing long sums like

$$x_1 + x_2 + \cdots + x_n$$

we can write the same in a more compact form:

$$\sum_{i=1}^n x_i .$$

This is read as *the sum of x_i from i equals one to n .*

There's a few components to this notation:

Examples

Example 1: Evaluate $\sum_{i=1}^5 i^2$.



Example 2: Evaluate $\sum_{n=1}^3 (2n + 1)$.



Examples

Example 1: Evaluate $\sum_{i=1}^5 i^2$.

$$\sum_{i=1}^5 i^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$$

Example 2: Evaluate $\sum_{n=1}^3 (2n + 1)$.

$$\sum_{n=1}^3 (2n + 1) = (2(1) + 1) + (2(2) + 1) + (2(3) + 1) = 3 + 5 + 7 = 15$$

Exercises

 Try these out individually first.

1) Expand and evaluate: $\sum_{i=3}^6 3i$

2) Expand and evaluate: $\sum_{i=2}^5 (i^2 - 1)$

3) Complete the following using sigma notation:

$$x_1 + x_2^2 + x_3^3 + \cdots + x_n^n = \sum$$

4) A climate model predicts daily high temperatures $\hat{y} = [22, 25, 20]$ °C. Observed values are $y = [24, 23, 21]$ °C. Calculate the mean squared error (MSE) using the formula:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2.$$

Solutions



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Solutions

1)

$$\begin{aligned}\sum_{i=3}^6 3i &= (3 \cdot 3) + (3 \cdot 4) + (3 \cdot 5) + (3 \cdot 6) \\ &= 3(3 + 4 + 5 + 6) \\ &= 54\end{aligned}$$

2)

$$\begin{aligned}\sum_{i=2}^5 (i^2 - 1) &= (4 - 1) + (9 - 1) + (16 - 1) + (25 - 1) \\ &= 3 + 8 + 15 + 24 \\ &= 50\end{aligned}$$

3)

$$x_1 + x_2^2 + \cdots + x_n^n = \sum_{i=1}^n x_i^i$$

4) The formula is $\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$. Compute each squared error first:

$$(\hat{y}_1 - y_1)^2 = (22 - 24)^2 = 4$$

$$(\hat{y}_2 - y_2)^2 = (25 - 23)^2 = 4$$

$$(\hat{y}_3 - y_3)^2 = (20 - 21)^2 = 1$$

Then apply the formula:

$$\text{MSE} = \frac{4 + 4 + 1}{3} = 3 \text{ } ^\circ\text{C}$$

Vectors and matrices

Linear algebra introduction

Linear algebra “branch of mathematics concerning linear equations” (Wikipedia), sometimes also described as the math of vectors & matrices.

It is a fundamental part of data science (and how computers understand & process data), and useful for describing environmental processes.

The building blocks of linear algebra



The building blocks of linear algebra

- **scalar:** a value without direction, representing magnitude. For our purposes, a number:

8 or λ .

- **vector:** an ordered list of values, representing magnitude *and* direction (physics) or values for an observation or variable (data science). For example,

$(1, 3, 5)$ or (x_1, x_2, x_3) .

- **matrix:** an array of values made up of rows and columns. For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 10 & 20 & 30 \end{pmatrix}$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

Applications of linear algebra in environmental sciences

- Dimensional reduction
- Population matrix models
- Optimization
- Array programming / vectorized code
- Machine learning

Let's start with vectors

Where are vectors in EDS?

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Where are vectors in EDS?

Everywhere.

Vectors are lists of values used to describe different features or variables of interest. For example, if you are trying to model *fish size* based on length (cm) and mass (g), then for a fish with length 32 cm weighing 281 g, you might describe that by:

$(32, 281)$

Sometimes vectors are represented with an arrow over the vector name:

$\vec{x} = (32, 281)$



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Vectors are lists of values used to describe different features or variables of interest. For example, if you are trying to model *fish size* based on length (cm) and mass (g), then for a fish with length 32 cm weighing 281 g, you might describe that by:

$$(32, 281)$$

Sometimes vectors are represented with an arrow over the vector name:

$$\vec{x} = (32, 281)$$

Often, we will think of an “abstract vector” x with n **coordinates**:

$$x = (x_1, x_2, \dots, x_n).$$

Vector addition & subtraction

Just add or subtract the corresponding coordinates.



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Just add or subtract the corresponding coordinates.

If: $u = (1, 2)$ and $v = (3, -1)$, then:

$$u + v = (1 + 3, 2 - 1) = (4, 1)$$

What does this look like graphically? **Let's draw it!**

Scalar multipliers

You can multiply any vector by a scalar (constant). This will not change the *direction* of the vector - it will only change the *magnitude* of the vector.



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Example: $u = (1, 2)$

$$w = 3u = (3 * 1, 3 * 2) = (3, 6)$$

What does this look like graphically? **Let's draw it!**

Vectors with > 3 coordinates

What about a vector with more than two coordinates? More than three?

Is as valid as describing a “point” in multivariate space as a vector with two “coordinates” – it’s just difficult for us to visualize and conceptualize since our brain only happily deals with 3 dimensions.

Dot product



Dot product

For vectors

$$x = (x_1, \dots, x_n) \text{ and } y = (y_1, \dots, y_n)$$

their **dot product** is:

$$x \cdot y = \sum_{i=1}^n x_i y_i$$

In words: The **dot product** is the sum of coordinates of each vector multiplied together. It is a measure of how close the vectors “point” in the same direction

Exercise

For vectors $x = (x_1, \dots, x_n)$ $y = (y_1, \dots, y_n)$ their **dot product** is:

$$x \cdot y = \sum_{i=1}^n x_i y_i$$

Find the dot product of $a = (2, -1, 0)$ and $b = (9, 3, -4)$:



Exercise

For $a = (2, -1, 0)$ and $b = (9, 3, -4)$:

$$a \cdot b = (2)(9) + (-1)(3) + (0)(-4) = 15$$

What happens when we have orthogonal vectors?



Sketch a quick graph, then find the dot product, of the following vector combinations:

1. $a = (0, 4)$ and $b = (6, 0)$

2. $x = (-3, 1)$ and $d = (2, 6)$

What is the value of the dot product for orthogonal vectors?

More on vector fundamentals:

Optional: watch 3Brown1Blue's great 10 min recording on [Vectors \(Ch 1 Essence of Linear Algebra\)](#).

Vector addition & scalar multiplication are the basis of most linear algebra!

Let's take a 5 minute break



image: Flaticon.com

Matrices

Matrices

A matrix is an array of values (has rows and columns). For example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}$$

- **Dimensions:** the size of the matrix, in rows x columns ($m \times n$)
- **Elements:** values in a matrix, often denoted symbolically with a subscript where the first number is the *row* and the second number is the *column* (e.g. a_{23} indicates the element in row 2, column 3)

Matrix algebra (add / subtract)

Add or subtract the corresponding elements (by matrix position) to create a new matrix of the same dimensions.



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Add or subtract the corresponding elements (by matrix position) to create a new matrix of the same dimensions.

$$\begin{bmatrix} 1 & -3 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} -1 & 4 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 5 & 8 \end{bmatrix}$$

Scalar multiplication

To multiply a matrix by a *scalar*, multiply each element in the matrix by the scalar to get a scaled matrix of the same dimensions.

For example:



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For example:

$$6 \times \begin{bmatrix} 1 & -3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -18 \\ 0 & 12 \end{bmatrix}$$

Recall: dot product

The dot product of two vectors is the sum of their elements multiplied:

For $u = (1, 5)$ and $v = (2, -3)$:

$$u \cdot v = (1)(2) + (5)(-3) = -13$$

Matrix multiplication

We find the *dot product* of row · column vectors:



Matrix multiplication

We find the *dot product* of row · column vectors:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \times \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{bmatrix}$$

Practice problems

 Find the product of these two matrices

$$\begin{bmatrix} 2 & -1 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 5 \\ 6 & -3 \end{bmatrix} = ?$$

Matrices with unequal dimensions

What would be the dimensions of the product if you were multiplying the following?

$$\begin{bmatrix} A & B & C \\ D & E & F \end{bmatrix} \times \begin{bmatrix} G & H \\ I & J \\ K & L \end{bmatrix} =$$

Let's try one!

$$\begin{bmatrix} 0 & 2 & -1 \\ 3 & 4 & -2 \end{bmatrix} \times \begin{bmatrix} -1 & 0 \\ 5 & 1 \\ -3 & 6 \end{bmatrix} = ?$$

Diagonal matrix

A **diagonal matrix** is (almost always) a square matrix ($m = n$) where only elements on the diagonal are non-zero values.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

What happens when we multiply a matrix by a diagonal matrix?

A diagonal matrix is also called a *scaling matrix* because it scales rows proportionally, but not by the same value:



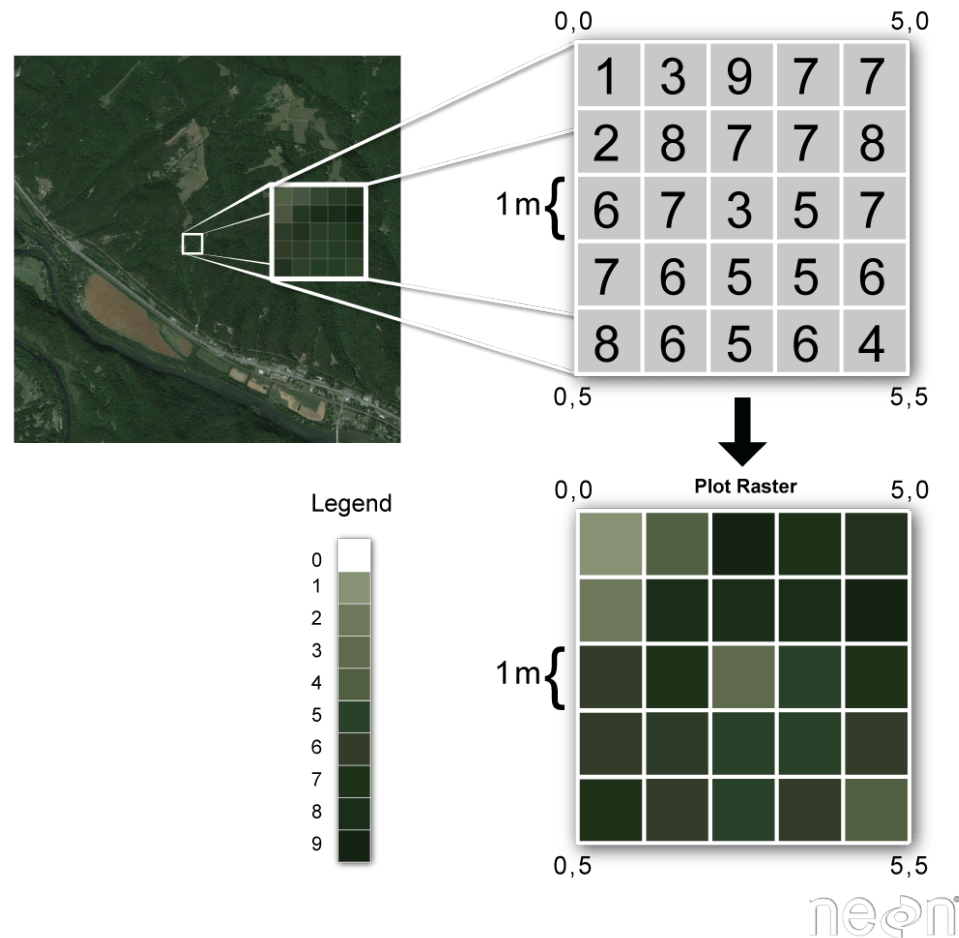
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A diagonal matrix is also called a *scaling matrix* because it scales rows proportionally, but not by the same value:

$$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \times \begin{bmatrix} 3 & 8 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 16 \\ 8 & 4 \end{bmatrix}$$

Where do matrices come up in EDS?

Rasters (matrix like)



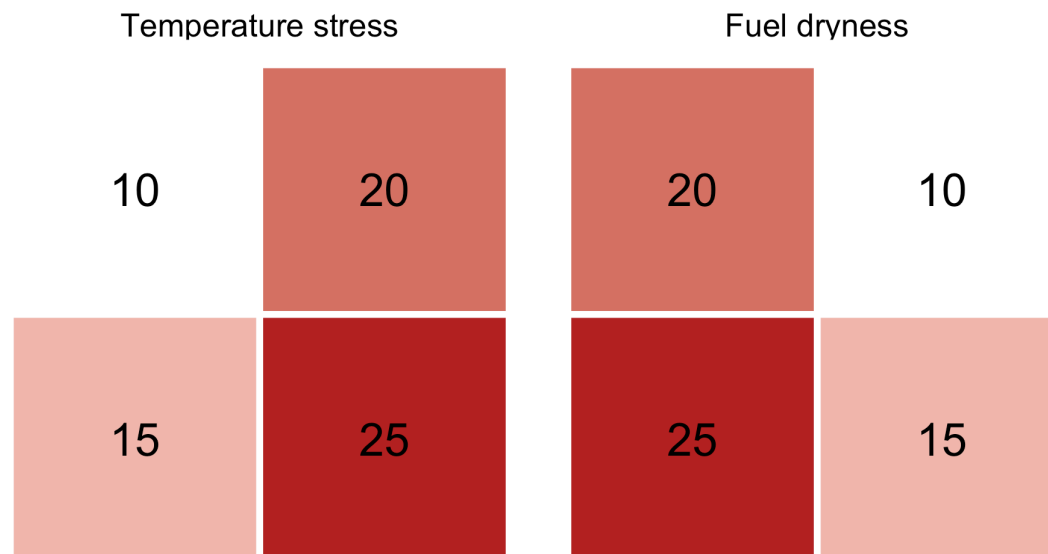
Source: Neon

Exercise: combining rasters

A raster is just a matrix where each cell holds a value for a location on a grid. Land managers often combine several raster layers into a single index by **scaling** each layer (multiplying by a weight) and **summing** the results.

 Calculate the fire risk index for each cell:

$$\text{Fire risk} = 0.6 \times \text{Temperature stress} + 0.4 \times \text{Fuel dryness}$$

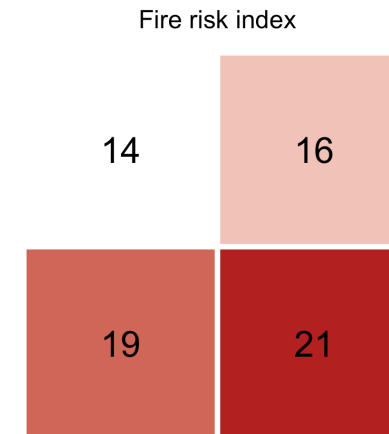


Exercise: solution

💡 Let's see a solution!

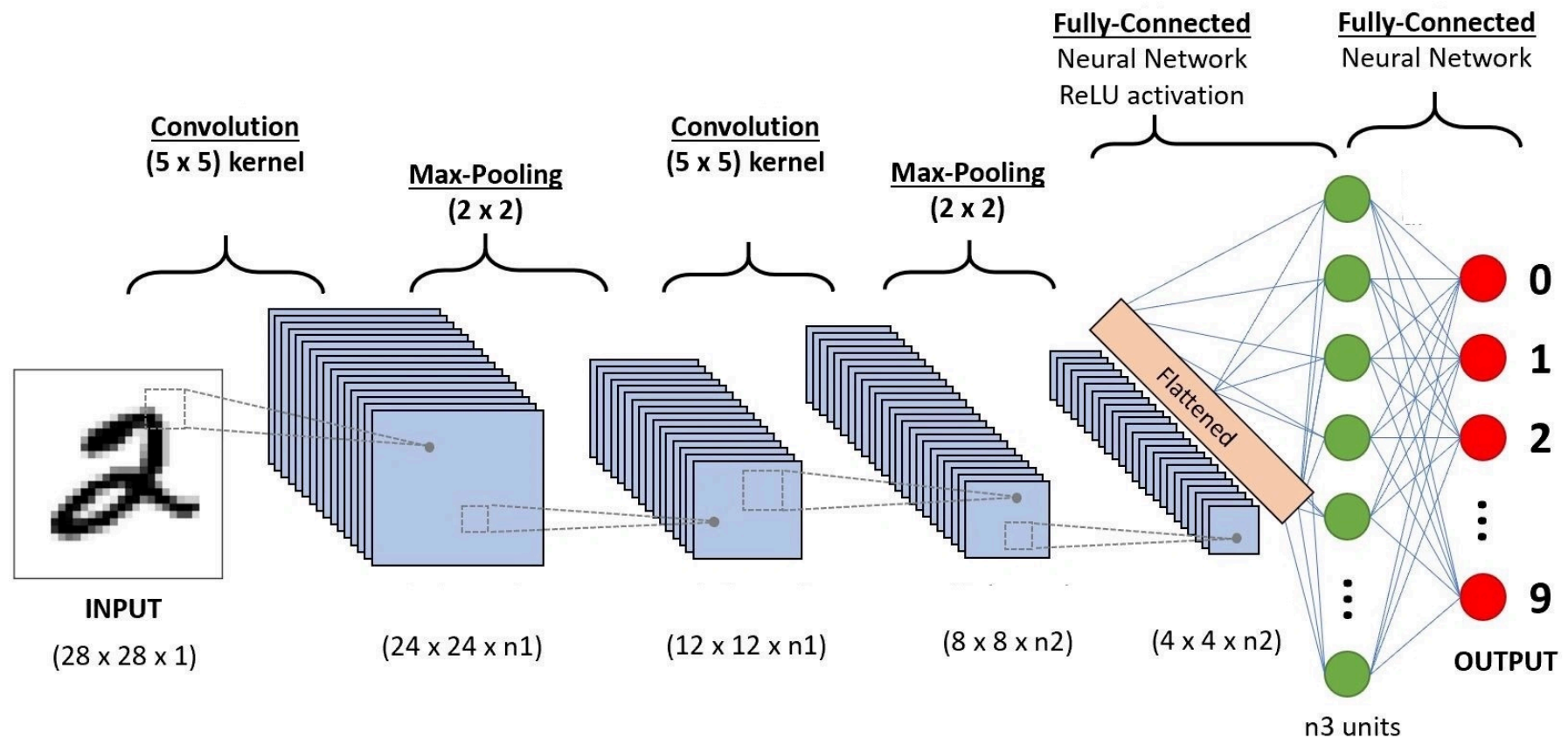
$$\text{Fire risk} = 0.6 \times \text{Temperature stress} + 0.4 \times \text{Fuel dryness}$$

- Top-left:
 $0.6(10) + 0.4(20) = 6 + 8 = 14$
- Top-right:
 $0.6(20) + 0.4(10) = 12 + 4 = 16$
- Bottom-left:
 $0.6(15) + 0.4(25) = 9 + 10 = 19$
- Bottom-right:
 $0.6(25) + 0.4(15) = 15 + 6 = 21$



Where do matrices come up in EDS?

Machine learning (EDS 232)



Adapted from [What is the Convolutional Neural Network Architecture?](#)

Wrapping up...

What we covered today

-  Practice all day 1 and 2 topics
-  A bit of ODEs
- $+$ Summation notation
-  Vector sum and product
-  Dot product
-  Matrix algebra